Bayesian Network Model for Safety Integrity Assessment

A Safety Instrumented Function Case Study

1 System Description

The system under analysis is a **Safety Instrumented Function (SIF)**, a fundamental building block of any **Safety Instrumented System (SIS)** as defined by the international standard **IEC 61508** (*Functional Safety of Electrical/Electronic/Programmable Electronic Safety-related Systems*) and its process-industry counterpart **IEC 61511**. This version of SIF is slightly personalized and different from a standard version, in particular it features a 2oo3 gate between sensors instead of an AND gate, and presents 3 sensors.

A SIF is an automated safety mechanism designed to bring a controlled industrial process to a safe state upon detection of a hazardous condition. It operates independently from the Basic Process Control System (BPCS) and activates only on demand — that is, when a measured process variable (e.g., pressure, temperature, or fluid level) exceeds a predefined safety threshold. A representative real-world deployment is an **overpressure protection loop** in an oil refinery or chemical plant: if the pressure inside a vessel rises beyond the safe operating limit, the SIF detects the anomaly and commands a shutdown valve to close, preventing a potential explosion or toxic release.

The functional chain of a SIF follows a fixed **Input–Logic–Output** architecture, and can fail at any of its three constituent subsystems. The objective of this study is to model the **failure behaviour** of such a system using a **Fault Tree Analysis (FTA)**, and subsequently to derive a **Bayesian Network (BN)** model that enables probabilistic inference and sensitivity analysis.

2 Dependability Attributes of Interest

Following the framework established by Avizienis et al. (2004), **dependability** is defined as the ability of a system to deliver service that can justifiably be trusted. The dependability attribute of interest for this study is **reliability** intended as **safety**, evaluated through two complementary quantitative metrics.

2.1 Probability of Failure in $[0, t]$

The primary quantitative measure adopted in this work is the **unreliability function**:

$$F(t) = 1 - R(t) = 1 - e^{-\lambda \cdot t}$$

where $R(t)$ is the reliability function and λ is the (constant) failure rate of the component, under the assumption of exponentially distributed failure times. This metric

directly answers the question: *"what is the probability that the system has failed at least once within a mission time t?"* For the system-level top event, $F_{top}(t)$ is obtained by propagating the component-level unreliabilities through the Fault Tree structure, and subsequently evaluated directly by the Bayesian Network model in Netica.

The reference mission time adopted in this study is $t = 8760$ h (one year), which corresponds to a standard proof test interval for industrial safety systems. The quantitative requirement for the SIF is:

$$F_{top}(8760) \leq 10^{-4}$$

That is, the probability that the SIF fails to respond to a demand within one year of operation must not exceed 10^{-4}.

2.2 Mean Time Between Failures (MTBF)

The secondary quantitative measure is the **Mean Time Between Failures (MTBF)** of the top event, defined as:

$$MTBF_{top} = \frac{1}{\lambda_{top}}$$

where λ_{top} is the failure rate of the top event, derived from the component failure rates through the Fault Tree structure. For small failure probabilities — which hold in the regime of interest ($\lambda_{top} \cdot t \ll 1$) — the following approximation holds:

$$\lambda_{top} \approx \frac{F_{top}(t)}{t}$$

allowing the MTBF to be computed directly from the Bayesian Network output:

$$MTBF_{top} \approx \frac{t}{F_{top}(t)}$$

This metric expresses the average time between hazardous failures of the SIF and provides an intuitive engineering interpretation of the system reliability. The quantitative requirement is:

$$MTBF_{top} \geq 10^4 \text{ years}$$

The two metrics are complementary: $F_{top}(t)$ is the primary output of the Bayesian Network analysis, while $MTBF_{top}$ is derived from it as a post-processing step. Both are reported and discussed in the results section.

3 System Architecture

The SIF modelled in this work consists of three subsystems connected in **functional series**: if any one subsystem fails to perform its role, the overall SIF fails. The three subsystems are:

1. **Sensor Subsystem** — detects the hazardous process condition and transmits a signal to the logic solver. This subsystem is implemented with **2oo3 (2-out-of-3) redundancy**: three independent sensors monitor the same process variable, and

the SIF fails at this stage if at least two sensors fail to operate correctly within the mission time, or if a Common Cause Failure (CCF) simultaneously affects enough sensors to defeat the voting logic. The use of 3 sensors is justified because it allows to understand which of the 3 is failed in the case of wrong measurements caused by a fault in a single sensor, this makes the system tolerant to 1 sensor that is sending altered data.

2. **Logic Solver Subsystem** — receives the sensor signal, evaluates the safety logic, and issues the actuation command. In practice this is implemented as a **safety-rated Programmable Logic Controller (PLC)**, certified in accordance with IEC 61508 Part 2.

3. **Actuator Subsystem** — executes the physical safety action in response to the logic solver command. In this model, the actuator is an **on/off pneumatic shutdown valve** whose failure mode of interest is *fail-to-close*: the valve remains open when it should close, allowing the hazardous condition to persist.

4 Failure Classification: DU and DD

A cornerstone of functional safety analysis under IEC 61508 is the classification of component failures along two orthogonal dimensions: their **effect on safety** and their **detectability**.

Effect on Safety

- **Safe failure**: the failure causes the system to move towards or into the safe state, even without a process demand. While this may cause a spurious trip (nuisance shutdown), it does not affect the reliability function $R(t)$ with respect to the hazardous top event and is therefore excluded from the Fault Tree.

- **Dangerous failure**: the failure *prevents* the SIF from responding correctly to a process demand, contributing directly to $F_{top}(t)$. This is the only failure category modelled in the Fault Tree.

Detectability

- **Detected (D)**: the failure is identified by the component's internal online diagnostics (e.g., watchdog circuits, signal range checks, internal self-tests). Upon detection, an alarm is raised and the operator can take corrective action. Detected dangerous failures (**DD**) are associated with a negligible residual contribution to $F_{top}(t)$ under the assumption of prompt maintenance response, and are therefore assigned a very small conditional probability in the model.

- **Undetected (U)**: the failure is *silent* — not revealed by online diagnostics. The component appears healthy but will not perform its safety function when demanded. Undetected dangerous failures (**DU**) accumulate over the mission time t and are the dominant contributor to $F_{top}(t)$. Their unreliability over the mission time is computed as:

$$F_{DU}(t) = 1 - e^{-\lambda_{DU} \cdot t} \approx \lambda_{DU} \cdot t \quad \text{for } \lambda_{DU} \cdot t \ll 1$$

The four resulting failure categories and their relevance to this study are summarised in Table 1.

Category	**Description**	**Dangerous?**	**Included in FT?**
Safe Detected (SD)	Visible to diagnostics	No	No
Safe Undetected (SU)	Hidden from diagnostics	No	No
Dangerous Detected (DD)	Revealed by diagnostics	Yes	Marginal
Dangerous Undetected (DU)	Hidden from diagnostics	Yes	**Yes**

Table 1: Failure mode classification according to IEC 61508.

5 Fault Tree Structure

The Fault Tree is rooted at the **Top Event**: *"SIF fails to respond to a process demand"*. This top event is decomposed using an **OR gate** into the three independent subsystem failure events, reflecting the series architecture of the SIF: any single subsystem failure is sufficient to cause overall SIF failure.

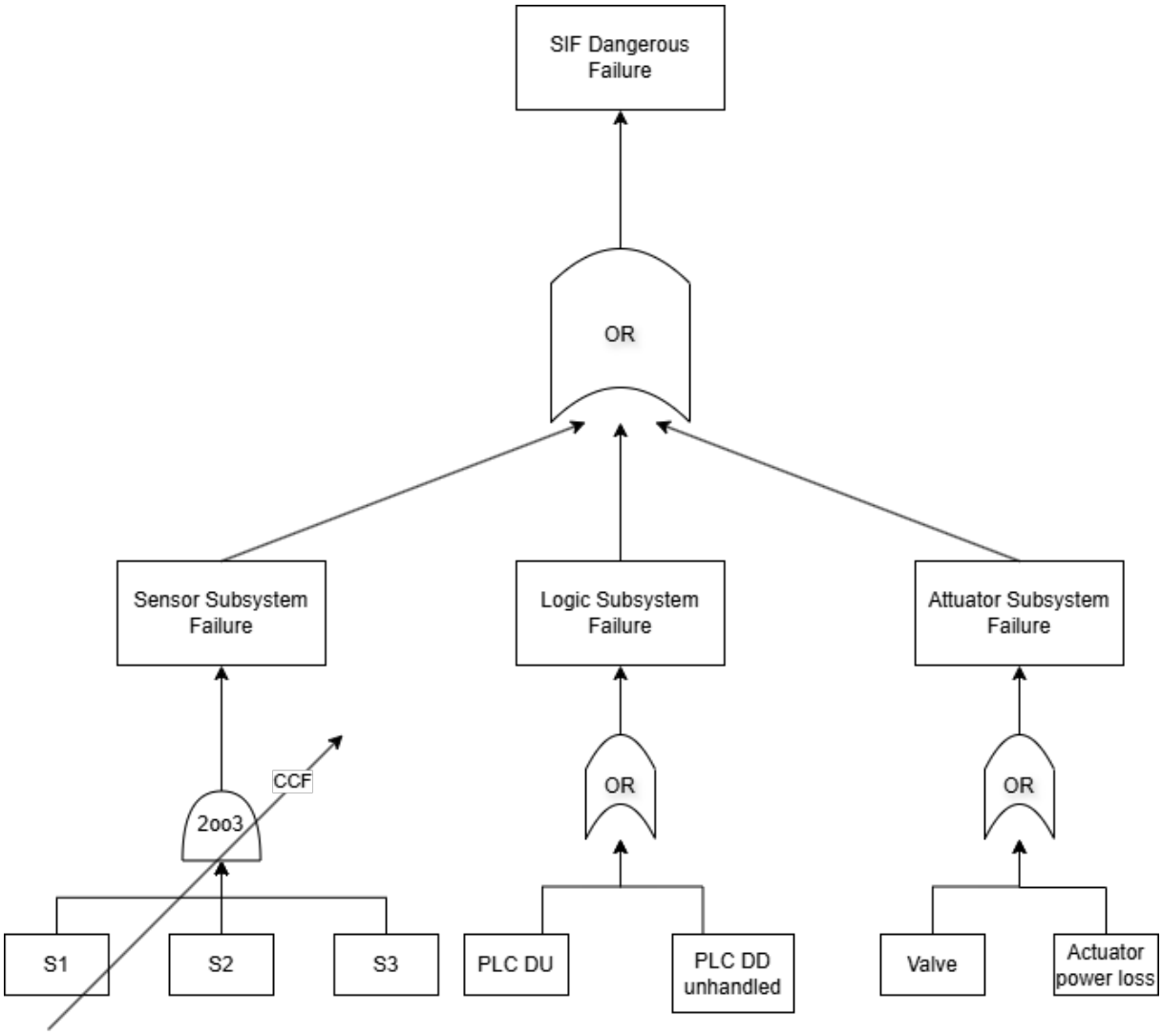


Figure 1: Fault Tree for SIF

Sensor Subsystem Failure

The sensor subsystem employs 2oo3 voting: the SIF loses its sensing capability if at least *two out of three* sensors fail. Two distinct causes can lead to this outcome:

1. **Independent voting failure**: at least two sensors fail independently within the same mission time t. Assuming identical and independent sensor unreliability $F_S(t)$, the probability of losing the sensing function is:

$$F_{2oo3,\,indep}(t) = \binom{3}{2} F_S(t)^2 \big(1 - F_S(t)\big) + F_S(t)^3$$

and for small probabilities $F_S(t) \ll 1$,

$$F_{2oo3,\,indep}(t) \approx 3F_S(t)^2$$

2. **Common Cause Failure** (`CCF_Sensors`): a shared environmental or systematic cause simultaneously incapacitates multiple sensors and defeats the 2oo3 vote. This

is modelled as a **probability**, graphically represented with the arrow crossing the 2oo3 gate, estimated via the **beta-factor model**:

$$\lambda_{CCF} = \beta \cdot \lambda_{DU}, \qquad F_{CCF}(t) = \beta \cdot \lambda_{DU} \cdot t$$

where $\beta \in [0.05, 0.10]$ for process sensors. The total sensor subsystem unreliability is then:

$$F_{sensor}(t) \approx 3F_S(t)^2 + \beta \cdot \lambda_{DU} \cdot t$$

Logic Solver Subsystem Failure

The logic solver (PLC) is modelled as a single component. Its unreliability over the mission time is:

$$F_{PLC}(t) = 1 - e^{-\lambda_{PLC,DU} \cdot t} \approx \lambda_{PLC,DU} \cdot t$$

The basic event `PLC_DU` accounts for hardware failures not covered by the internal diagnostics of the PLC. The detected failure branch (`PLC_DD`) is included for completeness with a negligible conditional probability, reflecting the assumption that detected failures are promptly handled by maintenance personnel.

Actuator Subsystem Failure

The actuator (shutdown valve) is modelled as a single component, with unreliability:

$$F_{valve}(t) = 1 - e^{-\lambda_{valve,DU} \cdot t} \approx \lambda_{valve,DU} \cdot t$$

The relevant failure mode is *fail-to-close on demand*, represented by the basic event `Valve_DU`. Common mechanical causes include spring fatigue, actuator stem seizure, and solenoid coil failure. The node **Actuator Power Loss** is included for completeness with a negligible conditional probability, to model for very rare occurrence of a power shortage to the valve mechanism.

6 Quantitative Parameters

The failure rates assigned to the basic events are drawn from established industrial data sources, including the **OREDA** handbook and **exida SERH** (Safety Equipment Reliability Handbook). The component-level unreliabilities $F_i(t)$ are computed at $t = 8760$ h and used directly as the prior probabilities of the basic event nodes in the Bayesian Network, assuming exponential failure rates.
Since detailed failure-rate data are not available for some secondary causes, such as **Actuator Power Loss** and **Unhandled Detected PLC Failures**, these nodes are modeled using small assumed prior probabilities. Their purpose is to preserve the causal structure of the BN rather than to provide an exact field estimate. Table 2 summarizes the baseline values.

Leaf node	Meaning	Rate / Assumption	Probability at $t = 8760$ h
S1	Sensor 1 DU failure	$\lambda = 1.0 \times 10^{-6}\,\mathrm{h}^{-1}$	0.00872
S2	Sensor 2 DU failure	$\lambda = 1.0 \times 10^{-6}\,\mathrm{h}^{-1}$	0.00872
S3	Sensor 3 DU failure	$\lambda = 1.0 \times 10^{-6}\,\mathrm{h}^{-1}$	0.00872
CCF_Sensors	Sensor common cause failure	$\beta = 0.10$	0.000876
PLC_DU	PLC DU failure	$\lambda = 5.0 \times 10^{-7}\,\mathrm{h}^{-1}$	0.00438
PLC_DD_unhandled	Unhandled PLC DD	$p = 5.0 \times 10^{-4}$	0.00050
Valve_DU	Valve DU failure	$\lambda = 2.0 \times 10^{-6}\,\mathrm{h}^{-1}$	0.01739
Actuator_Power_Loss	Actuator power loss	$p = 1.0 \times 10^{-3}$	0.00100

Table 2: Leaf-node probabilities at mission time $t = 8760$ h.

7 Bayesian Network Model

The Fault Tree introduced in the previous section has been translated into a Bayesian Network (BN) following the general principles proposed by Bobbio et al. for mapping fault trees into Bayesian networks. In particular, the BN preserves the causal structure of the original FT while replacing logical gates with probabilistic nodes and conditional probability tables (CPTs), thereby enabling probabilistic inference, evidence propagation, and sensitivity analysis.

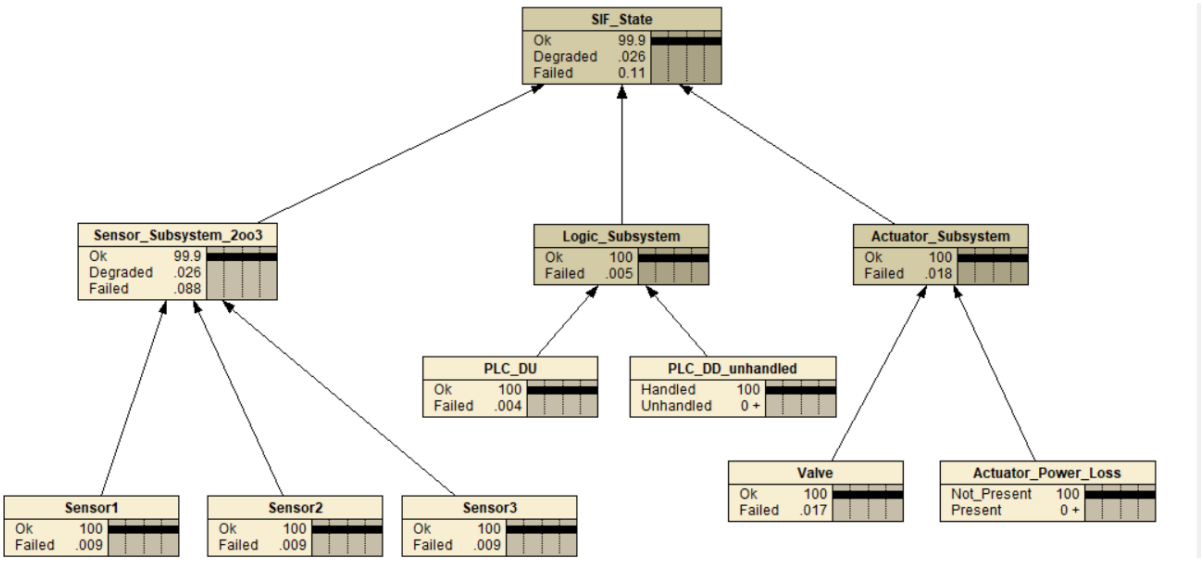


Figure 2: Bayesian Network

7.1 Network Structure

The resulting BN is composed of a set of binary leaf nodes, intermediate subsystem nodes, and a multistate top node representing the overall safety condition of the SIF. The leaf nodes correspond to the basic failure events identified in the FT, namely the three sensor dangerous-undetected failures (`S1`, `S2`, `S3`), the PLC dangerous-undetected failure (`PLC_DU`), the unhandled detected PLC failure (`PLC_DD_unhandled`), the valve dangerous-undetected failure (`Valve_DU`), and the actuator power-loss event (`Actuator_Power_Loss`). These basic events feed three intermediate subsystem nodes:

1. `Sensor_Subsystem`
2. `Logic_Subsystem`
3. `Actuator_Subsystem`

The **sensor branch** is modeled through a single CPT associated with `Sensor_Subsystem`, which directly maps the states of the three sensors to the states of the sensor subsystem. This formulation captures the 2oo3 voting logic together with the possibility of common-cause effects within the same conditional model, without introducing an additional explicit node in the graphical structure.
The **logic and actuator branches** are modeled in an analogous way, with OR-type combinations of their respective basic causes.
At the **top level**, the node `SIF_State` is modeled as a multistate variable with three outcomes, namely *Ok*, *Degraded*, and *Failed*. This choice makes the BN richer than the original FT, because it allows the model to distinguish between complete loss of the safety function and intermediate degraded conditions.

7.2 Differences with Respect to the Fault Tree

Compared to the original FT, the BN introduces two main modeling refinements. First, the top event is no longer represented as a purely binary event, but as a multistate node. This makes it possible to express not only the hazardous failure of the SIF, but also degraded operating conditions that do not immediately lead to loss of the safety function. Second, the sensor branch is represented by a probabilistic subsystem node whose CPT encodes both the voting behavior and the influence of common-cause mechanisms.

In the FT, the structure is limited to a deductive representation of the undesired event and is therefore mainly suited to computing the probability of the top event. In contrast, the BN retains the same causal dependencies but also supports diagnostic reasoning: once the top state is observed, the posterior probabilities of the basic events and subsystem nodes can be updated accordingly. This is one of the main advantages of the BN formalism over the FT approach, as discussed in the classical literature on FT-to-BN translation.

7.3 Assessment of the Reliability Target

The quantitative target adopted in this work, namely $F_{\text{top}}(8760\,\text{h}) \leq 10^{-4}$, represents a very stringent reliability requirement. Such a threshold is consistent with high-integrity safety functions operating in low-demand mode, where the probability of failure on demand must remain extremely small over the mission time. From an engineering perspective, this choice is reasonable for a Safety Instrumented Function (SIF) intended to provide a strong protective barrier against hazardous events.

The results obtained from the Bayesian Network indicate that the estimated probability of the top event failure over the one-year mission time is 0.11%, corresponding to $F_{\text{top}}(8760\,\text{h}) = 1.1 \times 10^{-3}$. Although this value is relatively low in absolute terms, it remains above the target threshold by approximately one order of magnitude. This suggests that the modeled system, in its present configuration, would still require additional risk reduction measures, such as improved component reliability, architectural redundancy, or shorter proof test intervals, in order to meet the specified safety objective.

Using the approximation $\lambda_{\text{top}} \approx F_{\text{top}}(t)/t$, the top-event failure rate can be estimated as

$$\lambda_{\text{top}} \approx \frac{1.1 \times 10^{-3}}{8760\,\text{h}} \approx 1.26 \times 10^{-7}\,\text{h}^{-1}.$$

The corresponding Mean Time Between Failures is then

$$MTBF_{\text{top}} \approx \frac{1}{\lambda_{\text{top}}} \approx 7.96 \times 10^{6}\,\text{h},$$

which is approximately 909 years. This confirms that, despite the non-negligible gap with respect to the target unreliability, the overall system failure frequency remains very low in absolute terms.

Therefore, the target can be regarded as realistic but demanding, and it provides a meaningful benchmark for evaluating the effectiveness of the proposed probabilistic model and the corresponding safety architecture.

8 Inference and Most Probable Explanation Analysis

The Bayesian Network (BN) was evaluated both through standard probabilistic propagation and through the Most Probable Explanation (MPE) functionality provided by Netica. These two inference modes offer complementary views of the model behavior. Standard propagation computes the posterior marginal distribution of each node given the imposed evidence, while MPE returns the single most probable joint assignment of the hidden variables consistent with that evidence.

8.1 Standard Posterior Propagation

In standard inference mode, evidence was imposed on the top node `SIF_State`, we don't cover the imposition on the *Ok* state because it just forces every component to be working. When the system state was set to *Degraded*, the posterior probabilities of the intermediate nodes and basic events were updated accordingly. The resulting distributions show that the sensor branch is the only contributor to the degraded top state, while the logic and actuator subsystems have to remain in the *Ok* state.

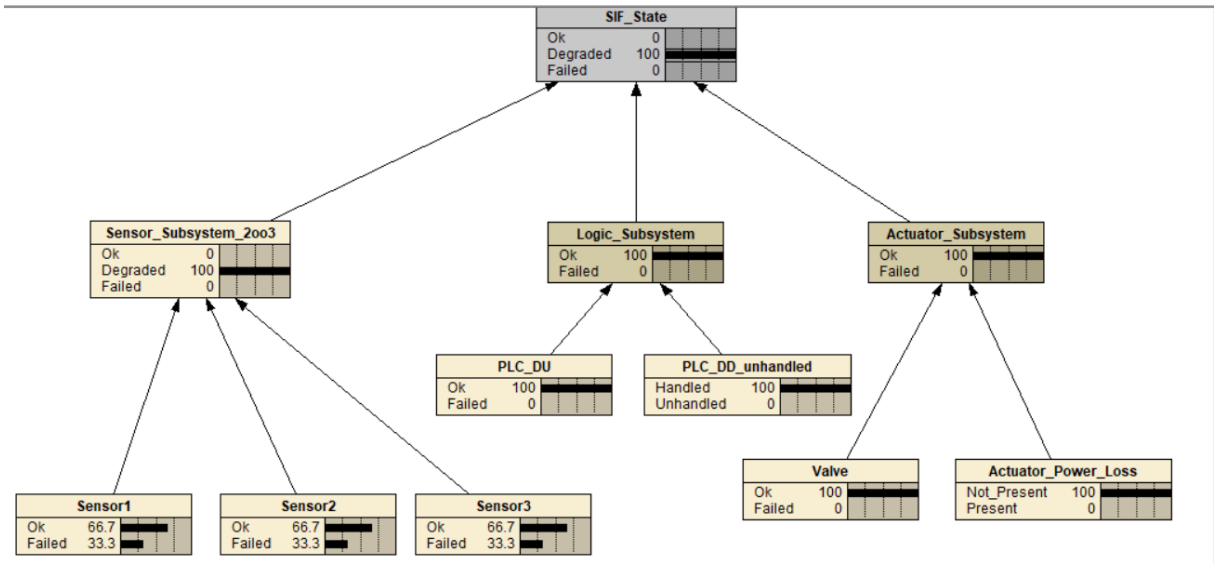


Figure 3: Bayesian Network with imposed Degraded state on the top event

In particular,as we can see in Figure 3 the node `Sensor_Subsystem_2oo3` becomes fully assigned to the *Degraded* state, whereas the three sensor basic events remain mostly in the *Ok* state with a non-negligible probability of failure of 33.3%, highlighting the fact that only one of the 3 has to be failed.

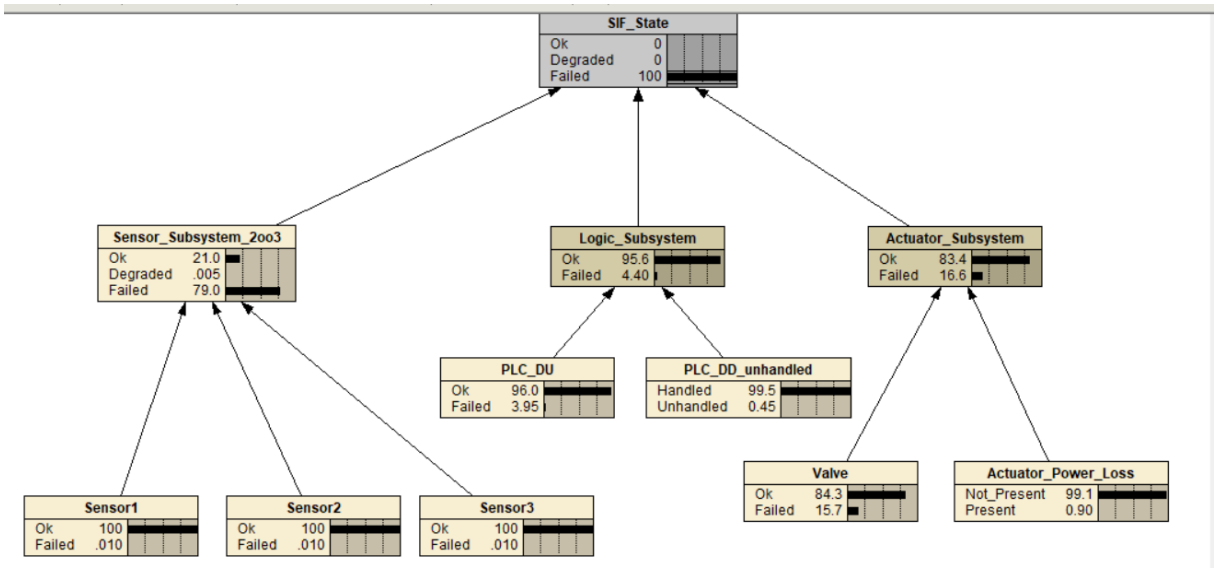


Figure 4: Bayesian Network with imposed Failed state on the top event

When the top node is set to *Failed*, as we can see in Figure 4, the posterior propagation still highlights a stronger involvement of the sensor branch. In this case, the probability mass concentrates on the *Failed* state of `Sensor_Subsystem_2oo3` (79%), while the logic and actuator subsystems show smaller but still non-zero failure probabilities (respectively 4.4% and 16.6%).
When we do not impose any condition, the Network shows the basic probabilities displayed

in Figure 2, with 0.11% probability of being in *Failed* and 0.024% of being in *Degraded*. This behavior is consistent with the structure of the BN, in which the top event is driven by the combination of the subsystem states.

8.2 Most Probable Explanation

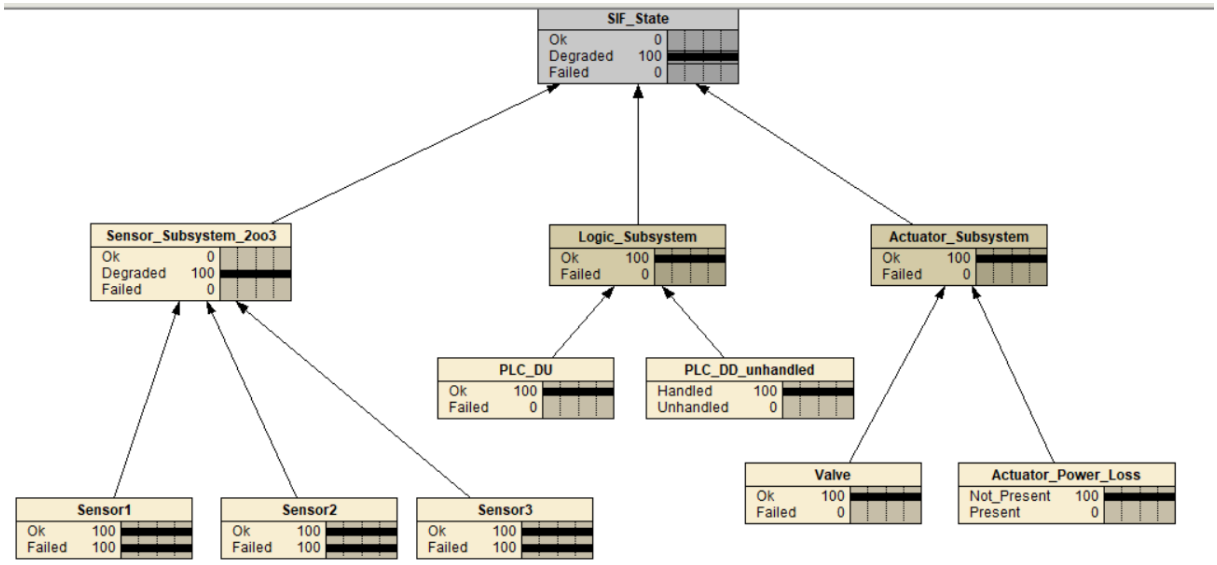


Figure 5: Bayesian Network under MPE analysis and with imposed Degraded state on top event

For the *Degraded* top state, the MPE returns the joint assignment with highest posterior probability under the constraint `SIF_State = Degraded`. In this configuration, the most likely explanation is of course the Sensor Subsystem having a failed sensor between the three, as we can see in Figure 5.

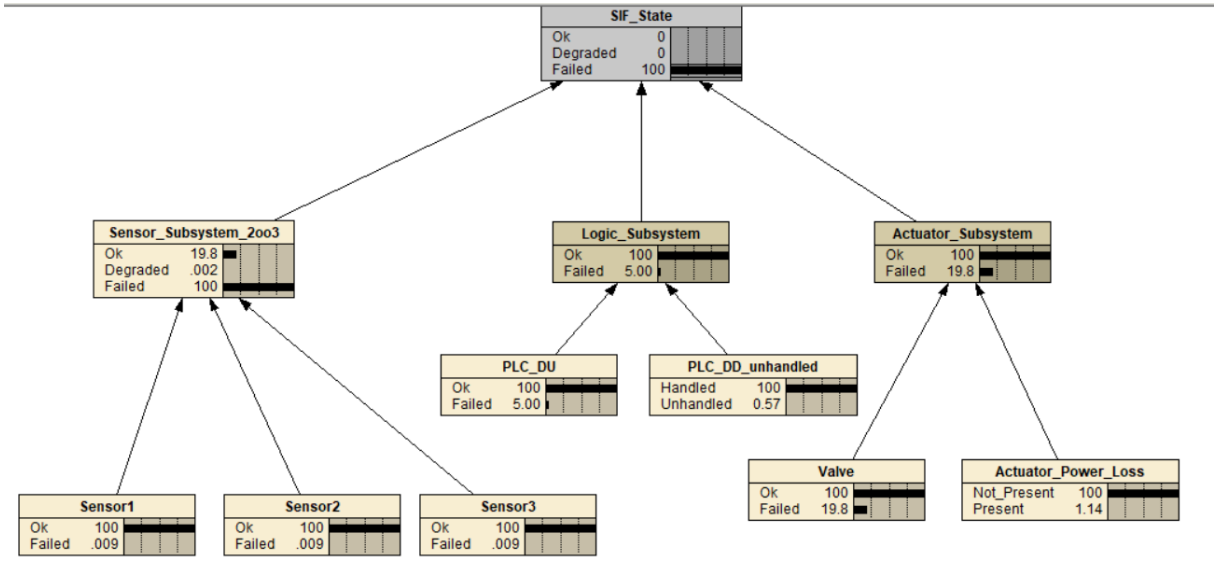


Figure 6: Bayesian Network under MPE analysis and with imposed Failed state on top event

For the *Failed* top state, the MPE yields an even more specific diagnosis. The sensor subsystem typically emerges as the most likely contributor to the failure, with `Sensor_Subsystem_2oo3` assigned to the *Failed* state in the most probable configuration, while still simngle sensor components get assigned to the most probable explanation of being *Ok* although the Sensor Subsystem saying otherwise.

8.3 Interpretation of the Results

In this model, the results indicate that the sensor branch is the most influential part of the SIF in both degraded and failed conditions. However, the interpretation depends on the inference mode used. Under standard propagation, multiple states may retain significant posterior probability. Under MPE, the analysis identifies one coherent system-wide explanation, which often corresponds to a dominant degradation or failure path.

Overall, the MPE functionality complements the posterior analysis by providing a deterministic diagnostic interpretation of the BN, while standard inference remains the preferred tool for quantifying uncertainty in the system behavior.